

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

CPE 314 — Past Questions & Materials
by Renance

Course: CPE 314
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 300

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Paper 1 of 1 — CPE 314 Assembly language programming 2025

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROL cng RESOURCES, EFFURUN

COLLEGE OF ENGINEERING any TECHNOLOGY

Examination: First Semester ~, Course Cade: CPE 314 (3 unit}

Session: 2024/2025 teat? Course Title; Assambly Language Programming

Date: /04/2025 Tims Allowed: 2hrs

Instruction: Answer Question ONE and any other FOUR Questions _

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QUESTION 4 (22 marks)

1. You have been required to produce an 8051-based (89C2051 or others) circuit to drive ten (10) LEDs

connected in acommon anode to a +5V source,

(i) Draw this circuit showing all details, (ii) Draw a flowchart that describes how you drive these LEDs, {iii)

Write the program that executes your flowchart, (iv) Submit your functional Circuit. {22 marks)

(a) By commenting on the Program code, interpret the function of each line and determine the operation

of the given program. Draw a flowchart for this program.

(b} Write an assembly language program for 8085 to check a hexadecimal number stored in 2500H memory location for odd or even parity. If the parity is even, store data EEH to memory location 2501H

Giherwise, store OOH in 2501H. (12 marks)

3. {a) Discuss the 8051 microcontroller with the Von Neuman architecture and show in a diagram how the

Address bus and Data bus connect all the various cornponents.

(b) What will be the value of the accumulator, CY and AC flags after the execution of the following program:

MVIA, 38H

ADI 87H

DAA

HLT (12 marks)

4, (a) With a diagram, outline the steps to Assembly language Programming from the source execution code in a microcontroller.

(b) Write an assembly language program for 8085 to find the 2's complement of the data stored in m code to

; emory

location 2100 H and the result is to be stored in memory location sete (12 marks)

5. (a) Convert the following binary numbers to their hexadecimal equivalent —

(i) 100111110101 (ii) 1001111011110101, and the following hexadecimal numbers to their

binary

form (iii) 29B (iv) 6A7D : ;

(b) Represent these data units as a power of 2 (i) 1 Kilobyte, (ii) 1 Megabyte,

(iii) 1 Gigabyte (iv) 1 Terabyte (12 marks)

Progra" SOunter ili) Input/output Registers

win its 2's compliment and ¢

agnitude:

6. a) Write short note on i) Accumulator ii)

b) Express the binary numbers given belo

etermine the decimal value of the

- . t i i oul

signed binary number expressed in sig"

01163510 ii) 11101000 ¢ 2 marks)

? \ of ASsemb} r

7. a} Explain the following —i) the Execution cv" ¥ Language programming ii)

Snstruction code) 2B + 84

Register Reference j

b} Add the following hexadecimal numbe!s- ; ii) DE + Ac

(12 marks)